MAKERERE UNIVERSITY

College of Computing and Information Sciences

School of Computing and Informatics Technology

Department of Networks

BSE4104 - Emerging Trends in Software Engineering

AI Native and Agentic Engineering Capstone Project

GROUP J EVENING

WEEK 1: PROBLEM FRAMING AND AI NATIVE REQUIREMENTS

SME PROCUREMENT PREPARATION AGENT

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PROJECT CHARTER

Problem statement

Small and medium sized enterprises in Uganda frequently experience stock outs or overstocking because procurement decisions are made manually under time pressure. Procurement officers or owner or managers must check inventory levels, decide reorder quantities, gather supplier quotations and prepare purchase requisitions. This process is time consuming, error prone and often lacks consistent documentation. The result is either lost sales due to stock outs or tied up capital due to excess inventory.

Target user

- Primary user: Procurement officer or owner or manager of a Ugandan small and medium enterprise.
- Secondary user: Designated approver such as owner or supervisor who must authorise any purchase requisition before it proceeds.

Current pain point

- Manual checking of stock against reorder points.
- Ad-hoc collection and comparison of supplier quotations.
- Time spent drafting requisitions that still require review.
- Lack of auditable trail showing why a particular quantity or supplier option was recommended.
- Risk of unauthorized or poorly justified purchases.

AI native solution and value proposition

The system is a bounded agentic application that assists the procurement officer through a controlled multi step workflow as below.

- Detect low stock items and recommend reorder quantities.
- Retrieve and compare current supplier quotations.
- Draft a complete purchase requisition with justification.
- Route the draft for mandatory human approval.

AI is used for reasoning, natural language explanation, quotation comparison narrative and draft generation. Deterministic software remains responsible for all numerical calculations, validation rules, stock arithmetic and enforcement of approval gates. The agent never

places orders, commits funds, selects a final supplier or sends communications without explicit human approval.

Assumptions and constraints

- All data will be synthetic and created by the team.
- The system will run on accessible foundation models API based with versioned prompts.
- High impact actions remain behind human approval gates at all times.
- The project must be fully demonstrable, testable and explainable within the eight week period.
- Group members remain accountable for system process through review, testing and documentation.
- We assume development will use an accessible foundation model which is to be selected and documented in Week 2 within realistic limits for our project.
- We assume all higher impact actions such as finalizing a requisition remain behind explicit human approval throughout the project as described by our integrity rules.

Success criteria

- The complete workflow can be executed end to end with synthetic data and produces a correctly formatted draft requisition.
- Every recommendation is grounded in retrieved inventory or quotation records.
- No high impact action can occur without recorded human approval.
- At least 30 evaluation scenarios are executed and scored.
- The final system is reproducible from a tagged GitHub release with clear run instructions.
- Every team member can explain the architecture, key design decisions, failures observed and residual limitations.